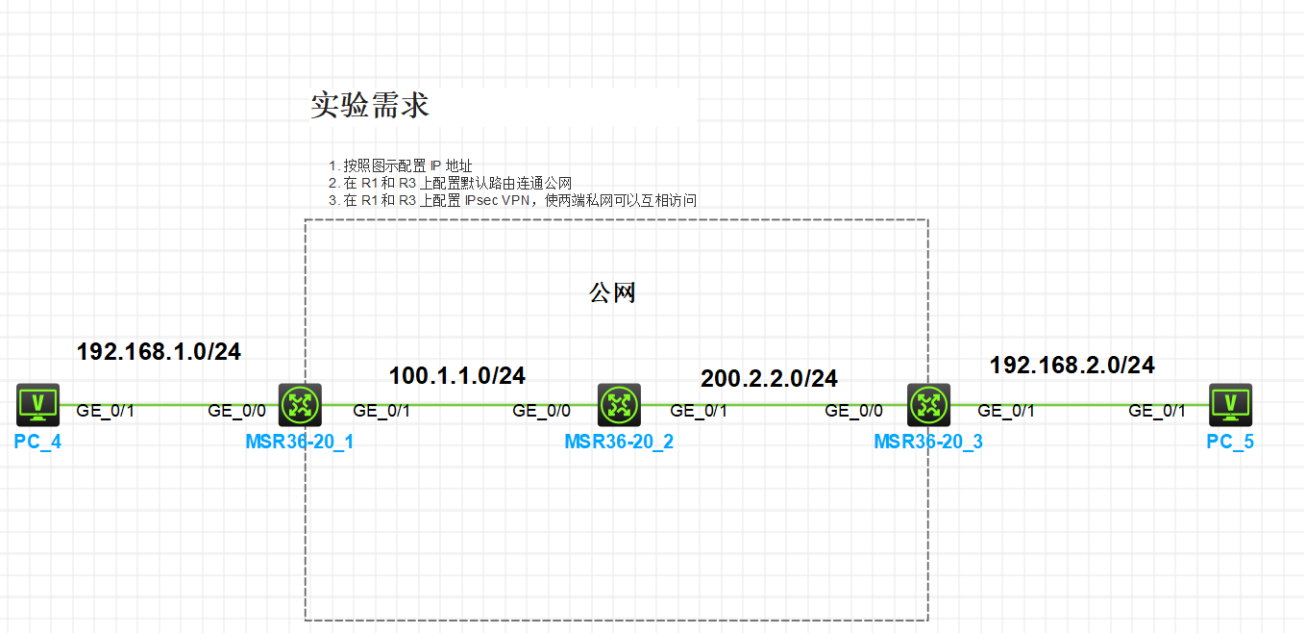


IP Sec VPN

实验拓扑



实验需求

1. 按照图示配置 IP 地址
2. 在 R1 和 R3 上配置默认路由由连通公网
3. 在 R1 和 R3 上配置 IPsec VPN，使两端私网可以互相访问

实验步骤

- 配置IP地址略
- 配置默认路由使公网互通

```
[R1]ip route-static 0.0.0.0 0 100.1.1.2

[r2]ip route-static 192.168.1.0 24 100.1.1.1
[r2]ip route-static 192.168.2.0 24 200.2.2.3

[R3]ip route-static 0.0.0.0 0 200.2.2.2
```

- 在 R1 和 R3 上配置 IPsec VPN，使两端私网可以互相访问

- 在 R1 上创建感兴趣流，匹配两端私网地址网段

```
[R1]acl advanced 3000
[R1-acl-ipv4-adv-3000]rule permit ip source 192.168.1.0 0.0.0.255 destination 192.168.2.0 0.0.0.255
[R1-acl-ipv4-adv-3000]dis this
#
acl advanced 3000
rule 0 permit ip source 192.168.1.0 0.0.0.255 destination 192.168.2.0 0.0.0.255
```

- 在 R1 上创建 IKE 提议，配置验证模式为预共享密钥，并配置加密算法

```
[R1]ike proposal 1
[R1-ike-proposal-1]authentication-method pre-share
[R1-ike-proposal-1]encryption-algorithm aes-cbc-128
[R1-ike-proposal-1]dis th
#
ike proposal 1
  encryption-algorithm aes-cbc-128
#
return
```

- 在 R1 上创建预共享密钥

```
[R1]ike keychain R3
[R1-ike-keychain-R3]pre-shared-key address 200.2.2.3 key simple 123456
[R1-ike-keychain-R3]dis th
#
ike keychain R3
  pre-shared-key address 200.2.2.3 255.255.255.255 key cipher
  $c$3$XcRzWzS1+e+ipaloXr8AciyvBLXtNy1bA==
#
return
```

- 在 R1 上创建 IKE Profile，指定本端和对端公网地址，并调用预共享密钥和 IKE 提议

```
[R1]ike profile R3
[R1-ike-profile-R3]keychain R3
[R1-ike-profile-R3]local-identity address 100.1.1.1
[R1-ike-profile-R3]match remote identity address 200.2.2.3
[R1-ike-profile-R3]proposal 1
[R1-ike-profile-R3]dis th
#
ike profile R3
  keychain R3
  local-identity address 100.1.1.1
  match remote identity address 200.2.2.3 255.255.255.255
  proposal 1
```

```
#  
return
```

- 在 R1 上创建IPsec 转换集，配置加密和验证算法。由于工作模式默认是隧道模式，且协议默认使用 ESP，所以无需配置

```
[R1]ipsec transform-set R3  
[R1-ipsec-transform-set-R3]esp authentication-algorithm sha1  
[R1-ipsec-transform-set-R3]esp encryption-algorithm aes-cbc-128  
[R1-ipsec-transform-set-R3]dis th  
#  
ipsec transform-set R3  
  esp encryption-algorithm aes-cbc-128  
  esp authentication-algorithm sha1  
#  
return
```

- 在 R1 上创建 IPsec 策略，调用上述配置

```
[R1]ipsec policy R3 1 isakmp  
[R1-ipsec-policy-isakmp-R3-1]security acl 3000  
[R1-ipsec-policy-isakmp-R3-1]ike-profile R3  
[R1-ipsec-policy-isakmp-R3-1]transform-set R3  
[R1-ipsec-policy-isakmp-R3-1]remote-address 200.2.2.3  
[R1-ipsec-policy-isakmp-R3-1]dis th  
#  
ipsec policy R3 1 isakmp  
  transform-set R3  
  security acl 3000  
  remote-address 200.2.2.3  
  ike-profile R3  
#  
return
```

- 在 R1 的公网接口上下发 IPsec 策略

```
[R1]int g0/1  
[R1-GigabitEthernet0/1]ipsec apply policy R3  
[R1-GigabitEthernet0/1]dis th  
#  
interface GigabitEthernet0/1  
  port link-mode route  
  combo enable copper  
  ip address 100.1.1.1 255.255.255.0  
  ipsec apply policy R3  
#  
return
```

- 在 R3 上完成 IPsec 相关配置，方法和命令与 R1 一致，本端和对端地址对调即可

```

[R3]acl advanced 3000
[R3-acl-ipv4-adv-3000]rule permit ip source 192.168.2.0 0.0.0.255 destination 19
2.168.1.0 0.0.0.255
[R3-acl-ipv4-adv-3000]dis th
#
acl advanced 3000
rule 0 permit ip source 192.168.2.0 0.0.0.255 destination 192.168.1.0 0.0.0.255
#
return
[R3]ike proposal 1
[R3-ike-proposal-1]authentication-method pre-share
[R3-ike-proposal-1]encryption-algorithm aes-cbc-128
[R3-ike-proposal-1]dis th
#
ike proposal 1
encryption-algorithm aes-cbc-128
#
return
[R3]ike keychain R1
[R3-ike-keychain-R1]pre-shared-key address 100.1.1.1 key simple 123456
[R3-ike-keychain-R1]dis th
#
ike keychain R1
pre-shared-key address 100.1.1.1 255.255.255.255 key cipher
$c$3$5dRQUiJS6ysmFk532Y9fcjZ7HNZT6fRkTw==
#
return
[R3]ike profile R1
[R3-ike-profile-R1]keychain R1
[R3-ike-profile-R1]local-identity address 200.2.2.3
[R3-ike-profile-R1]match remote identity address 100.1.1.1
[R3-ike-profile-R1]proposal 1
[R3-ike-profile-R1]dis th
#
ike profile R1
keychain R1
local-identity address 200.2.2.3
match remote identity address 100.1.1.1 255.255.255.255
proposal 1
#
return
[R3]ipsec transform-set R1
[R3-ipsec-transform-set-R1]esp authentication-algorithm sha1
[R3-ipsec-transform-set-R1]esp encryption-algorithm aes-cbc-128
[R3-ipsec-transform-set-R1]dis th
#
ipsec transform-set R1
esp encryption-algorithm aes-cbc-128
esp authentication-algorithm sha1
#
return
[R3]ipsec policy R1 1 isakmp
[R3-ipsec-policy-isakmp-R1-1]security acl 3000

```

```

[R3-ipsec-policy-isakmp-R1-1]ike-profile R1
[R3-ipsec-policy-isakmp-R1-1]transform-set R1
[R3-ipsec-policy-isakmp-R1-1]remote-address 100.1.1.1
[R3-ipsec-policy-isakmp-R1-1]DIS TH
#
ipsec policy R1 1 isakmp
transform-set R1
security acl 3000
remote-address 100.1.1.1
ike-profile R1
#
return
[R3]int g0/0
[R3-GigabitEthernet0/0]ipsec apply policy R1

```

○ 配置查看

```

[r1]dis ike sa

```

Connection-ID	Local	Remote	Flag	DOI
1	100.1.1.1	200.2.2.3	RD	IPsec

```

Flags:
RD--READY RL--REPLACED FD-FADING RK-REKEY
[r1]dis ipsec sa
-----
Interface: GigabitEthernet0/1
-----

-----
IPsec policy: r3
Sequence number: 1
Mode: ISAKMP
-----

Tunnel id: 0
Encapsulation mode: tunnel
Perfect Forward Secrecy:
Inside VPN:
Extended Sequence Numbers enable: N
Traffic Flow Confidentiality enable: N
Transmitting entity: Initiator
Path MTU: 1428
Tunnel:
    local address: 100.1.1.1
    remote address: 200.2.2.3
Flow:
    sour addr: 192.168.1.0/255.255.255.0 port: 0 protocol: ip
    dest addr: 192.168.2.0/255.255.255.0 port: 0 protocol: ip

[Inbound ESP SAs]
SPI: 4112849789 (0xf5251b7d)
Connection ID: 4294967296

Transform set: ESP-ENCRYPT-AES-CBC-128 ESP-AUTH-SHA1

```

```
SA duration (kilobytes/sec): 1843200/3600
SA remaining duration (kilobytes/sec): 1843199/3416
Max received sequence-number: 9
Anti-replay check enable: Y
Anti-replay window size: 64
UDP encapsulation used for NAT traversal: N
Status: Active
```

[Outbound ESP SAs]

```
SPI: 1866107230 (0x6f3a895e)
Connection ID: 4294967297
Transform set: ESP-ENCRYPT-AES-CBC-128 ESP-AUTH-SHA1
SA duration (kilobytes/sec): 1843200/3600
SA remaining duration (kilobytes/sec): 1843199/3416
Max sent sequence-number: 9
UDP encapsulation used for NAT traversal: N
Status: Active
```

[r1]dis ipsec policy

```
-----
IPsec Policy: r3
Interface: GigabitEthernet0/1
-----
```

```
-----
Sequence number: 1
Mode: ISAKMP
-----

Traffic Flow Confidentiality: Disabled
Security data flow: 3000
Selector mode: standard
Local address:
Remote address: 200.2.2.3
Remote address switchback mode: Disabled
Transform set: r3
IKE profile: r3
IKEv2 profile:
smart-link policy:
SA trigger mode: Traffic-based
SA duration(time based): 3600 seconds
SA duration(traffic based): 1843200 kilobytes
SA soft-duration buffer(time based): --
SA soft-duration buffer(traffic based): --
SA idle time: --
SA df-bit:
```

- 效果测试：在 PC4 上 Ping PC5，可以直接 Ping 通

```
<H3C>ping 192.168.2.1
Ping 192.168.2.1 (192.168.2.1): 56 data bytes, press CTRL_C to break
Request time out
56 bytes from 192.168.2.1: icmp_seq=1 ttl=253 time=1.769 ms
56 bytes from 192.168.2.1: icmp_seq=2 ttl=253 time=1.029 ms
56 bytes from 192.168.2.1: icmp_seq=3 ttl=253 time=1.243 ms
56 bytes from 192.168.2.1: icmp_seq=4 ttl=253 time=1.385 ms
```